

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (C.E./IT)

December 2012

CE-704 Soft Computing (SC)

Date: 21.12.2012, Friday Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. What is soft computing? How the hybridization of various technologies can be done? What are the applications of soft computing? 4
2. How can the equation of a straight line be formed using linear separability? Justify XOR function is non-linearly separable by a single decision boundary line. 4
3. Write and Give Genetic Algorithm encoding representation as an example for one specific Problem. 3

Q-2

- [A] In what ways is bipolar representation better than binary representation? State the training algorithm used for the Hebb network. 4

OR

- [A] What is Adaline? Draw the model of an Adaline network. Explain the training algorithm used in Adaline network. 4

- [B] What is the activation function used in radial basis function network? Explain the training algorithm of radial basis function network. 4

- [C] Indicate the two main types of associative memory. State the advantages of associative memory. Discuss the limitations of associative memory network. 4

OR

- [C] What is a Hopfield net? Compare and contrast BAM and Hopfield network. Mention the applications of Hopfield network. 4

Q-3

- [A] List the fixed weight competitive nets. What is winner takes all or clustering principle or competitive learning? 4

OR

- [A] With neat architecture explain the training algorithm of Kohonen self-organizing feature maps. Discuss the important features of Kohonen self-organizing maps. State the application of Kohonen self-organizing maps? 4

- [B] Explain Fuzzy Inference System in detail with its block diagram. 4

- [C] Define and explain in brief the Encoding System in GA. Write and Give the Encoding representation as an example for one specific Problem. 4

OR

- [C] Describe perceptron model in detail. What are the advantages of non linear activation function used in the simple perceptron? 4

SECTION-II

Q-4

1. What is the significance of Fuzzy Logic in Soft Computing? Discuss the advantages of fuzzy logic over Crisp Logic. Describe at least three characteristics to classify birds. Define fuzzy set for each characteristic and give example of birds that can be classified using these sets. 5
2. Explain the steps in the solution of a general optimization problem by a NN. What are the distinctive characteristics of multi layer perceptron? 3
3. Why a single layer of perceptron cannot be used to solve linearly inseparable problems? 3

Q-5

- [A] Where and how the knowledge is stored in NN? How do we decide how many hidden layers and number of neurons in each hidden layer should be there? "Increasing the number of hidden layer neurons gives a better function approximation". Is this statement correct? Justify your answer. 4
- [B] Why Genetic Algorithm is interesting? Why does it work? What are the advantages and disadvantages of Genetic Algorithms? 4
- [C] Which are the factors influencing the Back Propagation training? Define delta rule and give its uses in Back Propagation algorithm. 4

OR

Q-5

- [A] The input to a Single neuron is 5.0, its weight is 3.5 and its bias is -2 then calculates. 4
1. What is net input to transfer function?
 2. What is the neuron output?
- [B] What do you mean by Fuzzy decision making? Explain Fuzzy Inference System in detail with its block diagram. 4
- [C] What is the use of threshold function? Describe the various activation functions that are employed and compare their merits and demerits. 4

Q-6

- [A] Differentiate between feed forward and feed -back networks. Differentiate between competitive learning and supervised learning. 4
- [B] Narrate the following concepts with respect to ANN: 4
- Learning Rate, Bias, Variance, Momentum, Generalization, Under-fitting and Over-fitting.

OR

- [B] When to use Genetic Algorithms? Compare Genetic Algorithms with other search procedures. What are the application areas of Genetic Algorithms? 4
- [C] Explain the following Soft Computing (Hybridization) Techniques with their significance and applications: 4
1. Neuro-Fuzzy Hybrid Intelligent System
 2. Neuro- Genetic-Fuzzy Hybrid Intelligent System

OR

- [C] Define Crossover Rate and Mutation Rate. Explain Roulette Wheel Selection with an example. How Genetic Algorithms play important role while integrating it with the other technologies? Explain in brief. 4